

Applied Statistical Analysis I

Statistical inference review

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 September 24, 2025

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Today's class

- Lecture Recap
- Exercises

Key concepts and terms

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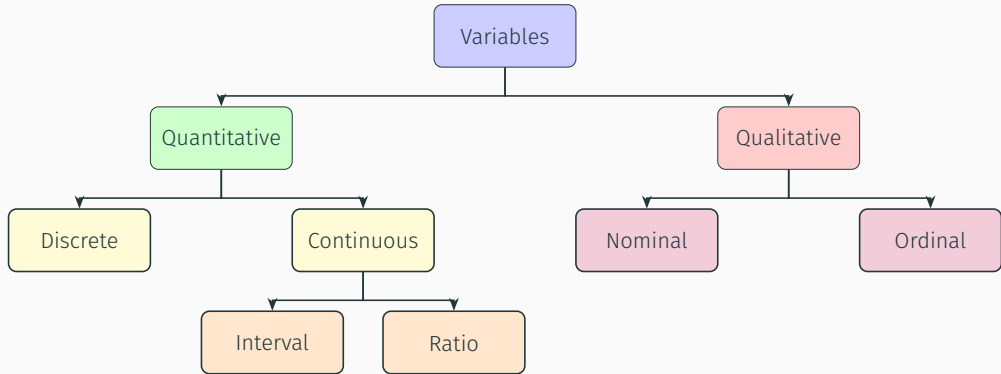
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- What is variation? (Example: Age $\rightarrow$ Income)
- *Necessary terms for regression analysis.*

Refers to the way variables are quantified. (e.g., economic wealth measured as GDP).

Measurement



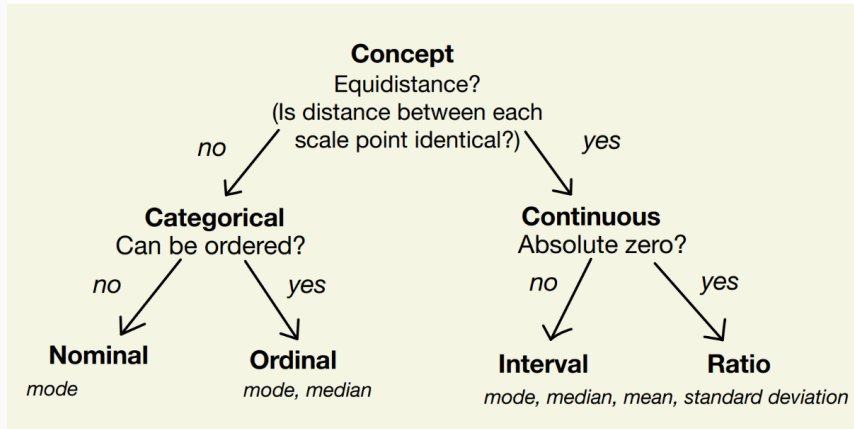
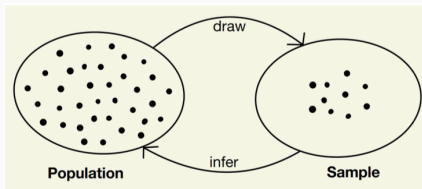


Figure 1: Kellstedt and Whitten 2018, Chap. 5

- What is the relationship between population and sample?

Population and Sample

- **Population:** the total set of subjects of interest in a study
- **Sample:** the subset of the population on which the study collects data
- **Parameter:** numerical summary of the population
- **Statistic:** a numerical summary of the sample data



- **Descriptive statistics:** “summarize the information in a collection of data”

Agresti and Finlay 2009, 4.

- **Inferential statistics:** “provide predictions about a population, based on data from a sample of that population”

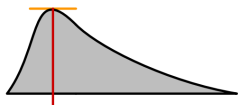
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Descriptive Statistics

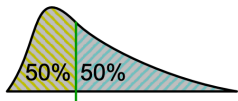
Measures of Central Tendency and Variability

- **Central tendency**: mean, median, mode
- **Variability**: variance, standard deviation, range, IQR
- **Visualization**: boxplots

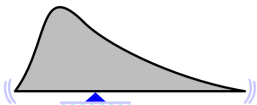
Descriptive Statistics: Central Tendency



mode



median



mean

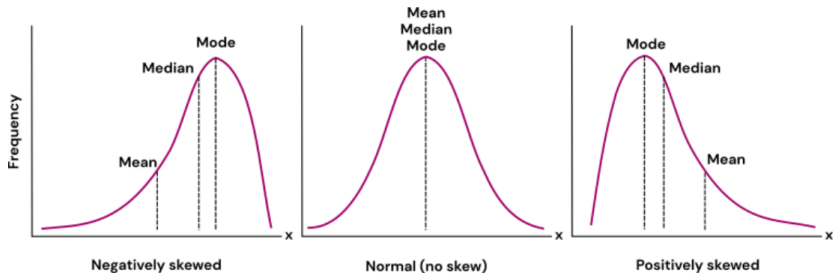
- **Mode**: Most frequently occurring value of X . Some distributions can have more than one mode.
- **Median**: Value of X that falls in the middle position when the observations are ordered from smallest to largest.
 - Median = 50th percentile = 2nd quartile
- **Mean**: Most common measure of central tendency.

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

Descriptive Statistics: Comparing Measures of Central Tendency

- In a perfectly symmetric distribution, e.g., normal distribution: **mode = median = mean**
- Not true when the distribution is non-symmetric:
 - right-skewed distribution (positive skew): $\text{median} < \text{mean}$
 - left-skewed distribution (negative skew): $\text{median} > \text{mean}$
- Mean is sensitive to outliers, while the median is more robust.

Descriptive Statistics: Comparing Measures of Central Tendency



Descriptive Statistics: Variability

- **Sample Variance**: Average of the square deviations from the mean:

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}$$

Why do we average by dividing by $n-1$? The sum of the deviations is always zero. Thus, the last deviation can be found once we know the other $n-1$. So we are not averaging n unrelated numbers. Only $n - 1$ squared deviations vary freely, these are called *degrees of freedom* of the variance.

- **(Sample) Standard Deviation**: Square-root of (sample) variance:

$$s = \sqrt{s^2} = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}}$$

- **Range**: Difference between largest and smallest measurement:

$$Range = x_{max} - x_{min}$$

- **Interquartile Range (IQR)**: Difference between upper and lower quartiles (range of the middle 50% of the distribution):

$$IQR = x_{Q3} - x_{Q1} \quad (1)$$

Probability

Probability: Basic terminology

- What is a probability?
- What is a distribution?
- What is a probability distribution?

Probability: Basic terminology

- An **experiment** is a repeatable procedure for making an observation.
- An **outcome** is a possible results of such an experiment.
- The **sample space** (Ω) of an experiment is the set of all possible outcomes.
- An **event** is a subset of the sample space, i.e., any set of outcomes.
- The **probability** of an event is its long-run relative frequency.
 - If $\Pr(A) = 0.5$, i.e., probability of event A is 0.5, then event A will occur approximately half of the time when the experiment is repeated infinitely often.
 - If the experiment is repeated many (finite) times, then the approximation as relative frequency (proportion) is expected to improve as the number of repetitions increases.

$$P(A) = \frac{\text{Number of elements in A}}{\text{Number of all elements}}$$

Probability: Distribution

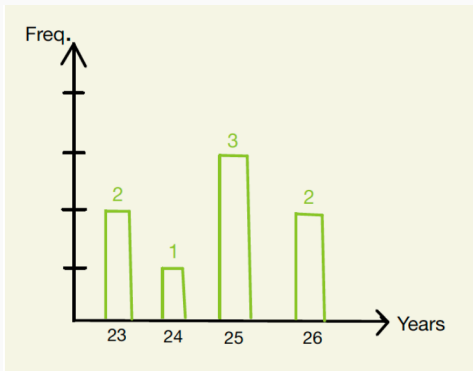


Figure 2: Example: Age of people in the room

It can take different shapes and, therefore, names: normal, binomial, t-distribution etc.

Probability: Probability Distribution

- Distributions of random variables are **probability distributions** if for all possible outcomes, it tells us the probabilities for these outcomes to occur.

Probability: Probability Distribution

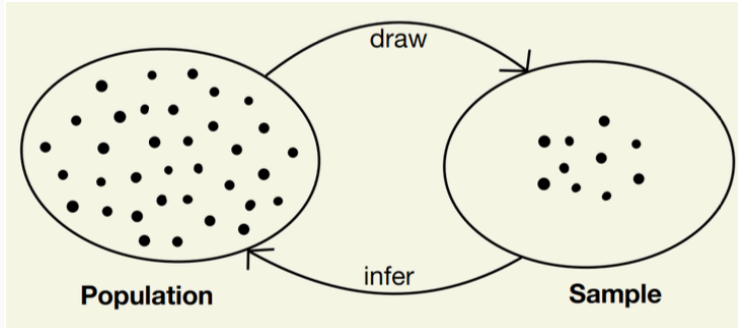
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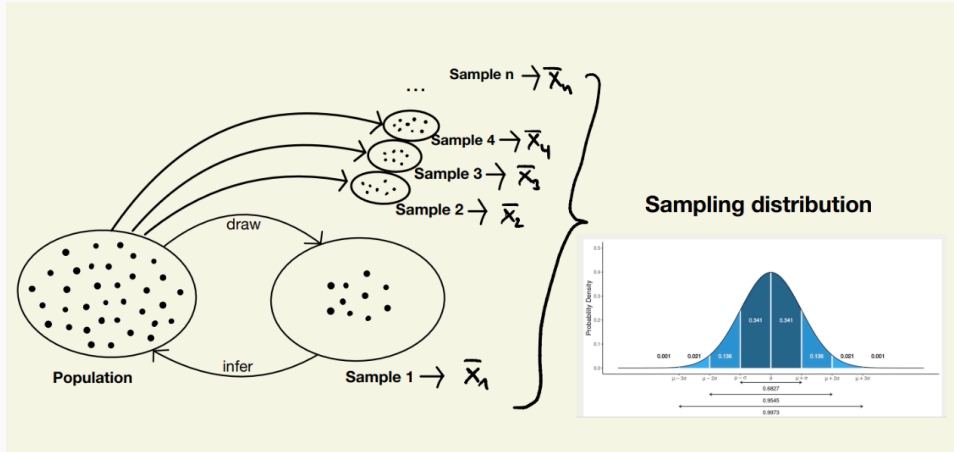
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- Definition by Agresti and Finlay (2009, 75): “lists the possible outcomes and their probabilities.”

Sampling Distribution

Recall the basic idea for empirical research:



Sampling Distribution



Sampling Distribution

- **Sampling Distribution:** “A sampling distribution of a statistic is the probability distribution that specifies probabilities for the possible values the statistic can take” (Agresti and Finlay, 2009, 87).
- For example: we have a population distribution with mean μ and variance σ^2 and we are interested in its mean.
- Repeatedly taking samples from that population and calculating the mean for each sample yields the **sampling distribution of the mean**.

Sampling Distribution

Why is this important?

- The corresponding probability theory “helps us predict how close a statistic falls to the parameter it estimates” Agresti and Finlay 2009, 87 → how close is $\bar{x}$ to μ ?
- Usually only one sample/estimate → **Point estimate**: “is a single number that is the best guess for the parameter value” Agresti and Finlay 2009, 107
- “If we repeatedly took samples, then in the long run, the mean of the sample means would equal the population mean μ ”.
- “The standard error describes how much $\bar{x}$ varies from sample to sample” → the SE is estimated based on the standard deviation:

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

Central Limit Theorem

What is the Central Limit Theorem?

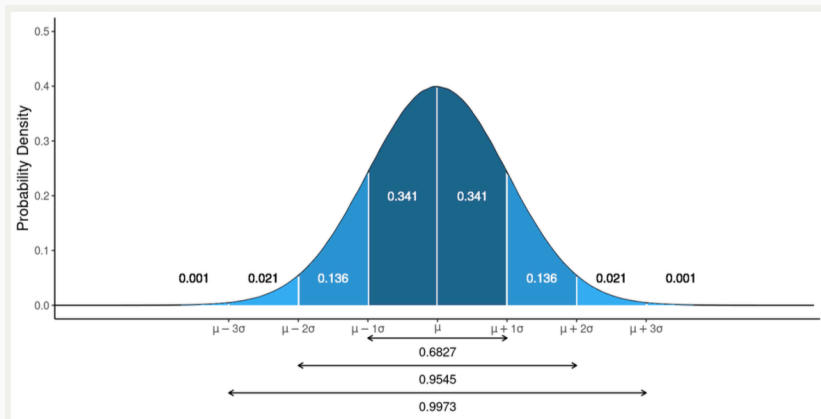
Central Limit Theorem

What is the Central Limit Theorem? → The sampling distribution of the statistic approaches a normal distribution with mean μ and variance σ^2/n as n increases.

- This holds *regardless* of the shape of the original population distribution.
- Basis for application of statistics to many 'natural' phenomena (which are the sum of many unobserved random events).
- How? Take a sample, calculate its mean. Do the same thing again and again.
The distribution of sample means will be normal even if the population distribution was not.
- If you repeatedly draw random samples from the same population, calculate the means and plot them, you get a histogram that approaches a bell-shaped curve.

Normal Distribution

- Continuous distribution that describes data clustered around the mean.
- Uniquely determined by its mean/median/mode μ and variance σ^2 .
- Important for the **Central Limit Theorem**.



Confidence Intervals

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Let's explain this a bit more!

Confidence Intervals

- Our estimate of a population parameter varies across repeated samples, thus generating a *sampling distribution*.
- Instead of a point estimate, we should better get an **interval estimate** - a **range within the true parameter lies with some level of certainty**.
- We can construct **confidence intervals** using the standard error or the variance of our estimates.
- We call a CI a q% confidence interval if it is constructed that it **contains the true parameter at least q% of the time if we repeat the experiment a large number of times**.
- Check out this visualization: <https://rpsychologist.com/d3/ci/>
- Attention! This does not mean that there is a q% probability for the population parameter to lie inside the interval!